

Internship

Semester – VIII

A PROJECT REPORT

Submitted by

Vadadoria Kaushal Manishbhai

210280107042

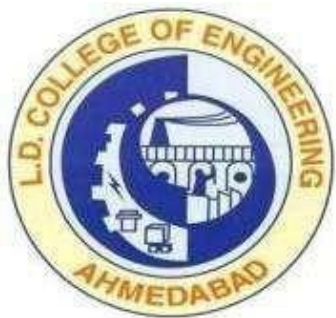
In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

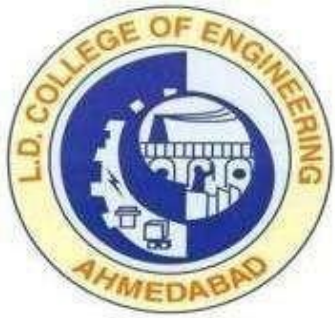
Computer Engineering

L.D. College of Engineering - Ahmedabad



Gujarat Technological University, Ahmedabad

April 2025



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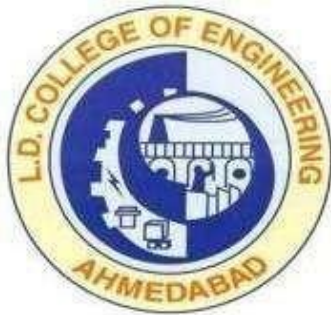
AHMEDABAD, GUJARAT

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **Internship** been carried out by **Kaushal Vadadoria** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2024-25.

Prof. Vimal Vaghela
Internal Guide

Prof. Chirag Thaker
Head of the Department



INTERNSHIP COMPLETION CERTIFICATE

 Business Solutions with Technology Edge	Document Name: Project Internship Certificate	
	Format Version No.: 1	Format Code: SSPL-HR-FR/01 Format Effective Date: 01-Apr-2024

April 14, 2025

To whom it may concern
L. D. College of Engineering,
Ahmedabad.

CERTIFICATE

This is to certify that **Mr. Kaushal Vadadoria**, currently a final semester student of the course **BE (CE)**, has successfully completed his internship project work titled "**Quiz Generator, AI-Powered QA Agent**" during the period from 10th January 2025 to 13th April 2025. He has completed his project work by using LangChain, LangGraph, Flask, ChromaDB

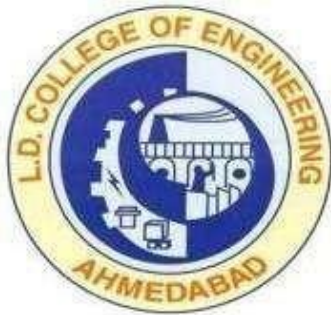
During his tenure of Project work, we have found his sincere, hardworking, dedicated, and punctual towards work. We wish him best of luck for all his future endeavors.

Thanking You,
Sarjen Systems Pvt. Ltd.

Chirag Prajapati
HR Department

Chirag
Prajapati

Digitally signed by Chirag Prajapati
DN: cn=Chirag Prajapati, o=Sarjen
Systems Pvt Ltd, ou=Human
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Date: 2025.04.14 15:08:29 +05'30'



INTERNSHIP JOINING LETTER

 Business Solutions with Technology Edge Format Version No.: 3	Document Name: Joining Intent Letter Format Code: SSPL-HR-FR/03/001 Format Effective Date: 19-Jul-2019	
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November 25, 2024,

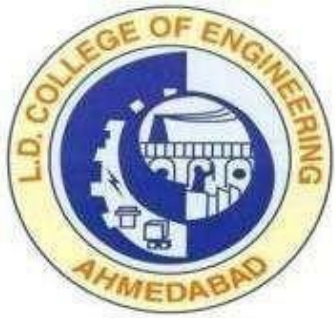
Mr. Kaushal Vadadoria,
LD - Ahmedabad,
Gujarat.

Subject: Joining Intent Letter

Dear Mr. Kaushal,

We are pleased to inform you that in reference to your resume submission and thereafter subsequent interview rounds, herewith we express our intent for employment.

Company	Sarjen Systems Pvt. Ltd.
Base Location	Ahmedabad
Team	Software Development
Role	Data Engineer - Generative AI / ML
Working Days	Monday to Friday
General Day Shift Office Timings	9.30 am – 6.40 pm Note: Above is the Head Office time. Projects are being executed in various shifts i.e. Indian Shift, UK Shift, US Shift. You will be allocated your shift as per your project.
Visits & Deputation	As per project's requirements OR as per client's invitation (Either International OR Domestic), only after attaining certain experience. Most of our projects are end to end implementation with leading clients mainly from pharmaceutical sector. As per requirements of projects, there will be opportunities to work at international/domestic client site, also in global projects shift (i.e. UK/US/Mexico/South East Asia/India etc).



L.D. COLLEGE OF ENGINEERING

AHMEDABAD, GUJARAT

DECLARATION

We hereby declare that the Internship / Project report submitted along with the Internship entitled **RAG Application and AI Agent** submitted in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me at **Sarjen Systems Pvt. Ltd.** under the supervision of **Prof. Vimal Vaghela** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Kaushal Vadadoria

Sign of Student

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to several individuals for supporting me throughout my internship project. First, I wish to express my sincere gratitude to my Internal Guide, **Prof. Vimal Vaghela**, for his enthusiasm, patience, insightful comments, helpful information, practical advice and unceasing ideas that have helped me tremendously at all times in my project. Without his support and guidance, this project would not have been possible.

I would like to express my endless thanks to my external guide **Mr. Rohan Ganeshrao Mundhe** at **Sarjen Systems Pvt. Ltd.** their sincere and dedicated guidance throughout the project development.

I am highly indebted to my institute guide **Prof. Vimal Vaghela** for the facilities provided to accomplish this internship. Also I would like to thank my Head of the Department **Pro. Chirag Thaker** for her constructive criticism throughout my internship. I am extremely grateful to my department staff members and friends who helped me in successful completion of this internship.

Thanking you,

Kaushal Vadadoria

210280107042

ABSTRACT

The **AI-Powered Quiz Generator** and **AI-Powered QA Agent** are advanced AI-driven systems designed to automate key processes in education and quality assurance.

The **AI-Powered Quiz Generator** leverages Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Natural Language Processing (NLP) techniques to automate quiz creation. The system processes input text, documents, or web content to extract key concepts and generate structured quiz questions, including multiple-choice, true/false, fill-in-the-blanks, and descriptive questions. By integrating LangChain for model interactions and using Ollama and Groq for on-device AI inference, the project ensures efficient and contextually accurate question generation. It incorporates semantic understanding, text preprocessing, and prompt engineering to generate well-balanced questions across various difficulty levels. A retrieval mechanism fetches reliable source materials to improve accuracy, while automated validation filters redundant or incorrect questions. The system also supports customization options for users, allowing them to select topics, difficulty levels, and question types, making the tool adaptable to different educational needs. Built with Streamlit for an intuitive web interface and deployable on Render for seamless accessibility, the tool reduces manual effort and ensures accurate and engaging assessments for educators, corporate trainers, and e-learning platforms.

The **AI-Powered QA Agent** focuses on **Deviation Detection** and **Corrective and Preventive Action (CAPA) Generation** in quality assurance processes. The system uses advanced machine learning and AI techniques to detect deviations in data, systems, or processes, flagging inconsistencies that deviate from established norms. Upon identifying these deviations, the system generates context-specific CAPAs, suggesting corrective actions and preventive measures to mitigate future occurrences. Integrated with real-time data sources and external systems, the QA agent provides valuable insights into operational processes, ensuring that issues are addressed promptly and effectively. By automating deviation detection and CAPA generation, this system reduces manual oversight, enhances compliance, and improves the overall quality of operations across industries such as manufacturing, healthcare, and finance.

Together, these AI-powered systems enhance efficiency in education and quality assurance. The **AI-Powered Quiz Generator** streamlines the quiz creation process, while the **AI-Powered QA Agent** improves operational quality through automated deviation detection and CAPA generation. Both systems combine state-of-the-art AI models, retrieval-based enhancements, and automation to offer scalable, intelligent, and reliable solutions for their respective domains.

List of Figures

Fig No.	Description	Page
Fig1.1	Company Logo	1
Fig 2.1	Production Stage	4
Fig 3.3	Gantt Chart	8
Fig 5.1	Flowchart	12
Fig 5.2	User Interface	13
Fig 10.1	Flowchart	28
Fig 10.2	User Interface	29

LIST OF TABLES

Table No.	Table Name	Page
Table 3.1	Project Summary (project -1)	5
Table 3.2	Technologies Used to Develop Project	7
Table 7.1	Test Cases	20
Table 8.1	Project Summary (project -2)	21
Table 8.2	Technologies Used to Develop Project	23
Table 12.1	Test Cases	36
Table 13.1	References	42

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
UI	User Interface
QA	Quality Assurance
CAPA	Corrective and Preventive Action
API	Application Programming Interface
ML	Machine Learning
CSV	Comma-Separated Values
CRUD	Create, Read, Update, Delete
IDE	Integrated Development Environment
UAT	User Acceptance Testing
TC	Test Case
UI/UX	User Interface / User Experience
GPU	Graphics Processing Unit

Table of Contents

Acknowledgement	i
Abstract	ii
List of Figures	iii
List of Tables	iv
List of Abbreviations	v
Table of Contents	vi
CHAPTER 1: OVERVIEW OF COMPANY	1
1.1 Company Introduction...	1
1.2 Different Services	1
CHAPTER 2: OVERVIEW OF COMPANY DEPARTMENT	3
2.1 All Departments...	3
2.2 Technical Specifications in Department...	3
2.3 Sequence of Operators.	4
2.4 Product Stages.....	4
CHAPTER 3: PROJECT 1: INTRODUCTION	4
3.1 Internship Project Summary.....	5
3.2 Purpose	6
3.3 Objective	6
3.4 Scope	6
3.5 Technology and Literature Review	7
3.6 Planning.....	7
3.6.1 Project/Internship Development Approach and Justification.....	7
3.6.2 Roles and Responsibilities.....	8
3.7 Scheduling.....	9
CHAPTER 4 PROJECT 1: SYSTEM ANALYSIS	9
4.1 Study Of Current System...	9
4.2 Weakness Of Current System...	9
4.3 Requirements of New System...	9

4.4 Features of New System.....	10
4.5 Selection of Hardware and Software.....	10
CHAPTER 5 PROJECT 1: SYSTEM DESING	12
5.1 Flowchart... ..	12
5.2 Interface Desing.....	13
CHAPTER 6 PROJECT 1: IMPLEMENTATION	18
6.1 Implementation Platform.....	18
CHAPTER 7 PROJECT 1: TESTING	19
7.1 Strategies	19
7.2 Testing Method... ..	20
7.2.1 Unit Testing.....	20
7.3 Testing Case.....	20
CHAPTER 8: Project 2: Introduction	21
8.1 Internship Project Summary.....	21
8.2 Purpose	22
8.3 Objective	22
8.4 Scope	22
8.5 Technology and Literature Review	23
8.6 Planning.....	23
8.6.1 Project/Internship Development Approach and Justification.....	23
8.6.2 Roles and Responsibilities.....	24
CHAPTER 9 PROJECT 2: SYSTEM ANALYSIS	25
9.1 Study Of Current System... ..	25
9.2 Weakness Of Current System... ..	25
9.3 Requirements of New System... ..	26
9.4 Features of New System.....	26

9.5 Selection of Hardware and Software.....	27
CHAPTER 10 PROJECT 2: SYSTEM DESING	28
10.1Flowchart... ..	28
10.2Interface Desing.....	29
CHAPTER 11 PROJECT 2: IMPLEMENTATION	34
11.1. Implementation Platform.....	34
CHAPTER 12 PROJECT 2: TESTING	35
12.1 Strategies.....	35
12.2 Testing Method... ..	36
12.2.1 Unit Testing.....	36
12.3 Testing Case	36
CHAPTER 13: CONCLUSION AND DISCUSSION	37
13.1 Overall Analysis Of Internship... ..	37
13.2 Problem Encountered and Possible Solutions... ..	37
13.3 Date Of Continuous Evaluation... ..	38
13.4 Limitations... ..	38
13.5 Future Enhancement.....	40
REFERENCES.....	42

CHAPTER 1 – OVERVIEW OF COMPANY



Figure1.1 Company Logo

1.1 Company Introduction

Sarjen Systems Pvt. Ltd., established in 1998, is a technology company specializing in IT consulting and software solutions across various industries, including pharmaceuticals, life sciences, and manufacturing. The company offers a diverse range of services such as enterprise mobility, analytics and AI, IoT and device interface, cloud integrations, customized business applications, and hyper-automation. With a client presence in over 50 countries, primarily in the UK and Europe, Sarjen has formed partnerships with major technology providers like Microsoft, Apple, Amazon, and Azure.

Sarjen has developed several proprietary solutions tailored to specific business needs. For instance, Process XE® is a Manufacturing Execution System (MES) designed to enhance manufacturing efficiency and quality while reducing errors and contamination risks. This system includes features like Electronic Batch Records (EBR), Weighing and Dispensing Management Systems (WDMS), and eLogbooks. Additionally, Sarjen offers solutions such as PharmaNET for supply chain and sales distribution management, FForce for sales force automation, QEdge for enterprise quality management, and PvEdge for pharmacovigilance safety databases.

The company's expertise spans various technologies, including web design, mobile development, programming languages, databases, and cloud infrastructures. Sarjen is committed to delivering business and GxP-compliant solutions, ensuring adherence to regulatory guidelines and industry standards. Their comprehensive approach to digital transformation and automation positions them as a trusted partner for organizations seeking to enhance operational efficiency and compliance.

1.2 Different Services

Sarjen Systems Pvt. Ltd. offers a comprehensive suite of services tailored to meet the diverse needs of industries such as pharmaceuticals, life sciences, FMCG, and manufacturing. Their key services include:

1. Enterprise Mobility Solutions: Sarjen develops mobile applications that enhance business operations, enabling seamless access to enterprise systems on-the-go. These solutions are designed to improve productivity and ensure data integrity across various devices.

2. Analytics and Artificial Intelligence (AI): Leveraging advanced analytics and AI technologies, Sarjen provides insights that drive informed decision-making. Their services encompass data modeling, predictive analytics, and machine learning applications tailored to specific business requirements.

3. Internet of Things (IoT) and Device Interface: Sarjen integrates IoT solutions to connect and manage devices, facilitating real-time data collection and analysis. This service aids in optimizing processes and enhancing operational efficiency across various industrial applications.

4. Cloud Integrations: Offering cloud integration services, Sarjen enables businesses to migrate, manage, and optimize applications in cloud environments. Their expertise ensures scalable, secure, and cost-effective cloud solutions aligned with organizational goals.

5. Customized Business Applications: Understanding unique business challenges, Sarjen designs and develops tailored applications that address specific operational needs. Their solutions aim to streamline processes, improve user experience, and achieve strategic objectives.

6. Hyper Automation: Sarjen implements hyper-automation strategies by combining robotic process automation (RPA), AI, and other technologies to automate complex business processes. This approach enhances efficiency, reduces errors, and accelerates digital transformation initiatives.

These services reflect Sarjen's commitment to delivering innovative and compliant solutions that drive business growth and operational excellence.

- Business Development
- Software Product Development
- Software Testing & QA

CHAPTER 2 – OVERVIEW OF THE COMPANY DEPARTMENT

Sarjen Systems Pvt. Ltd., established in 1998, is a technology company specializing in IT consulting and software solutions across various industries, including pharmaceuticals, life sciences, and manufacturing. The company offers a diverse range of services such as enterprise mobility, analytics and AI, IoT and device interface, cloud integrations, customized business applications, and hyper-automation. With a client presence in over 50 countries, primarily in the UK and Europe, Sarjen has formed partnerships with major technology providers like Microsoft, Apple, Amazon, and Azure.

2.1 All Departments

- **Human Resources (HR) Department**
- **Development Department**
- **Quality Assurance (QA) and Compliance Department**

2.2 Technical Specifications in Departments

- **Human Resources (HR) Department**

The HR team at Sarjen Systems is dedicated to attracting, developing, and retaining top talent. They manage recruitment, compensation, training, and development programs to foster a positive and productive work environment. Their robust management practices create an informal group of like-minded individuals working with zeal and a unified spirit, bringing fun to the workplace.

- **Sales and Marketing Department**

Sarjen's sales and marketing professionals work diligently to understand market dynamics and develop strategies that effectively reach target audiences. They focus on positioning Sarjen's diverse range of solutions, which transform business and technology operations to world-class excellence. Their in-depth knowledge of various technology areas enables them to provide end-to-end solutions tailored to client needs.

- **Development Department**

The development team at Sarjen comprises skilled professionals specializing in various technologies, including Microsoft .NET. They focus on creating high-quality, user-friendly applications tailored to clients' specific needs. The team's expertise spans web design, mobile development, programming languages, databases, and cloud infrastructures.

- **Quality Assurance (QA) and Compliance Department**

Sarjen's QA and Compliance department ensures that all software solutions meet stringent quality standards and regulatory requirements. They implement processes like Deviations, CAPA, Change Controls, and Risk Assessments to maintain compliance, particularly for clients in the pharmaceutical and life sciences sectors. Their commitment to quality is reflected in their ISO 9001:2015 certification, emphasizing development and implementation of enterprise software applications.

2.3 Sequence of Operations

1. **Requirement Gathering & Analysis:** The initial phase involves comprehensive discussions with clients to gather and analyze requirements. The team defines the features to be developed and estimates the project's cost and timeline, ensuring alignment with client expectations.
2. **Development & Team Allocation:** Based on the project requirements, specialized teams are allocated to handle different aspects of development. For instance, the ASP.NET team may manage backend development, while the UI/UX team focuses on creating intuitive interfaces. This tailored approach ensures that each project component is handled by experts in the respective fields.
3. **Testing & Quality Assurance:** Upon completion of development, the QA and Compliance department conducts rigorous testing to identify and rectify any issues. This meticulous process ensures that the software meets all functional and regulatory requirements before deployment.

Sarjen Systems' structured approach to software development, combined with their commitment to quality and compliance, positions them as a trusted partner for organizations seeking to enhance operational efficiency and achieve digital transformation.

2.4 PRODUCT STAGES



Fig 2.1 Production Stage

CHAPTER 3 PROJECT 1: INTRODUCTION

3.1 PROJECT SUMMARY

Table 3.1: Project Summary

Project Titles	Quiz Generator	
Aim	The aim of the AI-Powered Quiz Generator is to automate quiz creation using AI, NLP, and LLMs, generating accurate and diverse questions from text and PDF. It enhances learning and assessment by supporting multiple question formats and ensuring real-time validation. This tool is designed for educators, trainers, and organizations to create quizzes efficiently, improving engagement and knowledge evaluation.	
Developed At	Sarjen Systems Pvt. Ltd.	
Project Category	Web app	
Tools	IDE	Visual Studio
	Languages/Frameworks	Python, LangCahin, Flask,
Duration	3 Months (January 2025 to April 2025)	

3.2 PURPOSE

The purpose of this AI-Powered Quiz Generator project is to **streamline the quiz creation process** by leveraging AI, NLP, and LLMs to generate high-quality questions from various sources. It aims to **reduce manual effort, enhance learning experiences, and provide accurate assessments** for educators, trainers, and organizations. By automating question generation, the project ensures **efficient, scalable, and adaptive** quiz creation for diverse educational and training needs.

3.3 OBJECTIVE

- 1 **Automate Quiz Creation** – Develop an AI-driven system that generates quiz questions from text, documents, or web content, reducing manual effort.
- 2 **Enhance Learning & Assessment** – Provide diverse question formats (MCQs, true/false, fill-in-the-blanks) to support effective knowledge evaluation.
- 3 **Ensure Accuracy & Relevance** – Utilize NLP and LLMs to generate contextually accurate and meaningful quiz questions.
- 4 **Increase Efficiency & Scalability** – Enable educators, trainers, and organizations to create quizzes quickly and at scale.
- 5 **Adaptive Questioning** – Implement AI-powered difficulty adjustment to match the learner's knowledge level.
- 6 **Improve Engagement** – Make quizzes more interactive and insightful by offering AI-generated explanations and feedback.

3.4 SCOPE

The AI-Powered Quiz Generator has a **wide scope** across various domains, including **education, corporate training, and online assessments**. It automates quiz creation using **AI, NLP, and LLMs**, allowing users to generate **contextually relevant and accurate questions** from diverse sources such as documents, textbooks, and web content.

The project can be implemented in **e-learning platforms, schools, universities, coaching institutes, and corporate training programs** to streamline assessments. It supports **multiple question formats (MCQs, true/false, fill-in-the-blanks, and descriptive questions)** and can **dynamically adjust difficulty levels** based on the learner's proficiency.

Future enhancements may include **multilingual support, integration with LMS platforms, AI-driven feedback mechanisms, and real-time quiz analytics**. The system can be expanded to support **interactive quizzes with gamification elements**, making learning more engaging and personalized.

3.5 TECHNOLOGY AND LITERATURE REVIEW

Front End	Html, css, js
Back End	LangChain, RAG, Flask

Table 3.2 Technologies Used to Develop Project

3.6 PLANNING

3.6.1 Project / Internship Development Approach and Justification

The **AI-Powered Quiz Generator** will follow an **Agile development approach** to ensure flexibility, continuous feedback, and iterative improvements. **Modular development** will break the project into independent components, such as AI models, web scraping, frontend, backend, and database, ensuring efficient development and easy maintenance.

A **user-centered design** will prioritize user-friendly interfaces and meet the specific needs of educators and trainers. Technologies like **LLMs (AI)** for question generation, **RAG** for data retrieval, and **cloud deployment** will ensure scalability, real-time content integration, and cost-efficiency. Regular **usability testing** and **feedback integration** will ensure the system is intuitive and meets user expectations.

This approach guarantees that the project remains adaptable, scalable, and aligns with user needs while leveraging modern AI and cloud technologies for performance and flexibility.

The chosen **Agile methodology**, **modular development approach**, and **user-centered design principles** will ensure that the AI-Powered Quiz Generator is developed efficiently, aligns with user needs, and is scalable and maintainable. The justification for selecting advanced AI technologies, RAG, and cloud deployment underscores the project's focus on innovation, performance, and user satisfaction. This approach provides flexibility for iterative improvements and effective risk management to ensure the project's success.

3.6.2 Roles and Responsibilities

1. Project Manager

- **Responsibilities:**
 - Oversee the entire project lifecycle, ensuring it stays on track, within budget, and meets deadlines.
 - Coordinate between stakeholders, team members, and clients.
 - Manage project risks, resolve issues, and ensure that all tasks are completed on time.
 - Monitor project progress and conduct regular reviews with the team.

2. AI/ML Engineer

- **Responsibilities:**
 - Design and develop AI models, including **LLMs (Large Language Models)**, for question generation.
 - Train, test, and fine-tune AI models using relevant datasets to improve question quality and relevance.
 - Integrate **NLP (Natural Language Processing)** techniques for better understanding of content and context.
 - Work with the development team to integrate AI functionalities into the backend

3.7 SCHEDULING

Week	2	3	4	5	6	7	8	9	10	11	12	13	14
Month Activity	January				February				March				April
Domain Understanding													
Future Analysis													
Learning Process													
Design													
Coding and Testing													
Documentation													
Final Documentation													

Fig 3.3 Gantt Chart

CHAPTER 4 PROJECT 1: SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

Currently, quiz generation for educational purposes relies heavily on manual effort, where educators create and customize quizzes based on their own knowledge and research. While there are some automated systems, they are often based on predefined question templates or static databases that don't consider current information or dynamic content. These systems also lack personalized question generation, which can make quizzes repetitive and not as engaging.

Existing systems may involve the following limitations:

- **Limited scope** for content sources and question diversity.
- **Manual question creation**, which is time-consuming and prone to errors.
- **Fixed question formats** with no customization for specific learning outcomes.
- **Limited integration with external data sources**, preventing up-to-date content.

The AI-Powered Quiz Generator aims to address these limitations by automating question creation, offering diverse and dynamic question types, and integrating external data sources for real-time content generation.

4.2 WEAKNESS OF CURRENT SYSTEM

The primary weaknesses of the current systems include:

1. **Manual Effort:** Teachers or educators have to spend considerable time manually creating quiz questions, resulting in inconsistent quality, errors, and a lack of variety.
2. **Lack of Personalization:** Current systems do not adapt to the learning needs of individual students or specific learning outcomes.
3. **Limited Content Variety:** Most systems offer a fixed set of questions or rely on outdated content, making quizzes less engaging and up-to-date.
4. **Inflexibility:** Existing systems lack customization in terms of quiz formatting, question types, and difficulty levels.
5. **Inefficient Integration:** Many systems are not integrated with modern technologies like web scraping and AI, limiting the diversity and relevance of the questions generated.

4.3. REQUIREMENT OF NEW SYSTEM

The new AI-powered quiz generator system should fulfil the following requirements:

1. **Automated Question Generation:** Generate relevant and diverse questions from various content sources, leveraging **AI** and **NLP** for natural language understanding.
2. **Personalization:** Adapt quizzes based on learner profiles and educational objectives, allowing for dynamic difficulty adjustments.
3. **Integration with External Data:** Fetch up-to-date content from external sources using web scraping and **Retrieval-Augmented Generation (RAG)** techniques to enhance the question pool.

4. **Customizable Quiz Creation:** Provide flexible options for educators to design quizzes with different question types (e.g., multiple-choice, short answer, true/false).
5. **User-Friendly Interface:** The system should be easy to use for educators, students, and administrators, with an intuitive front-end for quiz creation, management, and results tracking.
6. **Data-Driven Insights:** Offer analytics and insights on quiz performance to help educators.

4.4. FEATURE OF NEW SYSTEM

Key features of the AI-Powered Quiz Generator:

1. **AI-Powered Question Generation:** Automatically generate context-relevant quiz questions using large language models and NLP.
2. **Dynamic Quiz Customization:** Allow educators to create customized quizzes based on difficulty, topic, and question types.
3. **Real-Time Content Integration:** Use web scraping and RAG to fetch up-to-date information, ensuring diverse and current quiz content.
4. **Adaptive Learning:** Adjust quiz difficulty and question relevance based on student progress and learning profiles.
5. **Analytics and Feedback:** Provide detailed insights into student performance, identifying strengths and areas for improvement.
6. **User-Friendly Interface:** Easy-to-use platform for educators and students, designed with intuitive navigation and options for personalization.
7. **Cloud-Based Infrastructure:** Ensure scalability, reliability, and cost-effectiveness by hosting the system on a cloud platform.

4.5. SELECTION OF HARDWARE AND SOFTWARE

- Hardware
 - Development Machine Requirement (Average performance)
 - 8 GB RAM
 - 50 GB HDD
 - Client Machine Requirement
 - 4 GB RAM
 - 5 GB of HDD space
 - Host Machine Requirement
 - 32 GB RAM
 - 500GB HDD Space

- Software
 - Development Machine Requirements
 - Visual Studio
 - Client Machine Requirements
 - Browser

CHAPTER 5 PROJECT 1: SYSTEM DESIGN

5.1. FLOWCHART

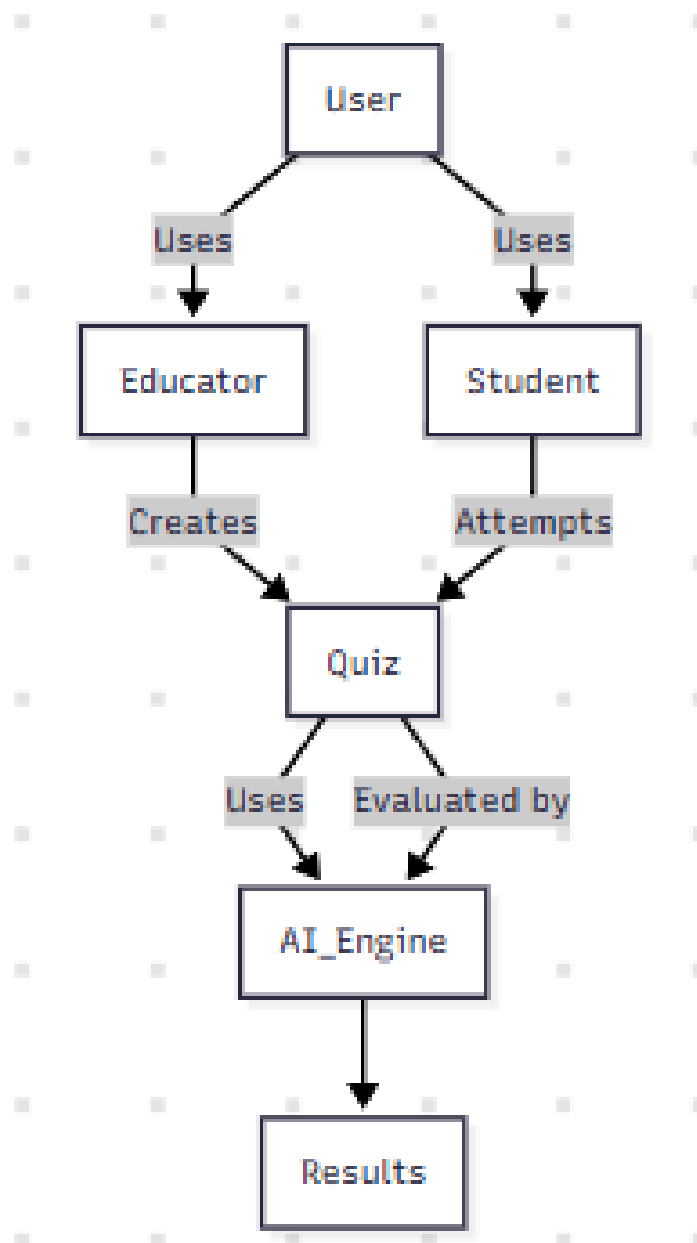
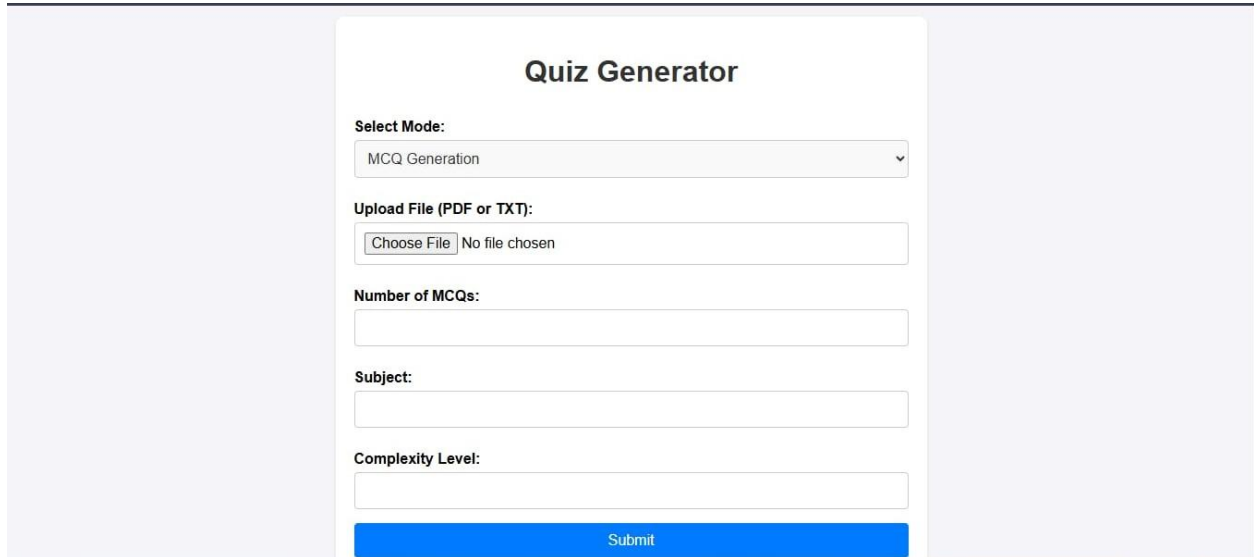


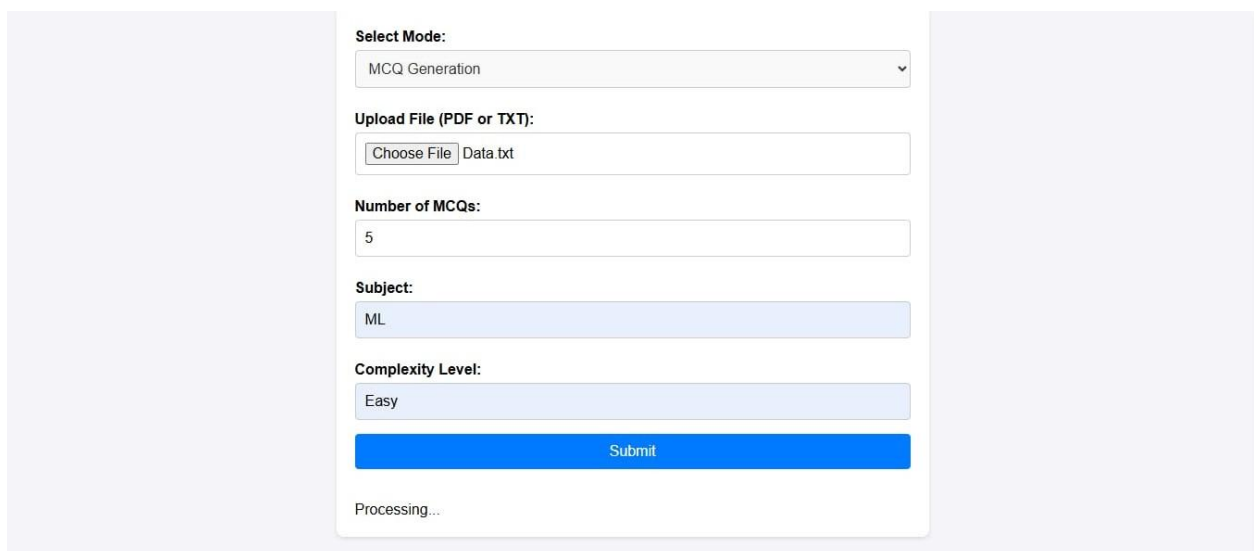
Fig 5.1 Flowchart

5.2. INTERFACE DESIGN



The screenshot shows a web interface titled "Quiz Generator". It features a dropdown menu for "Select Mode" with "MCQ Generation" selected. Below this is a file upload section labeled "Upload File (PDF or TXT)" with a "Choose File" button and the text "No file chosen". There are three text input fields: "Number of MCQs:", "Subject:", and "Complexity Level:". At the bottom is a blue "Submit" button.

Fig 5.2 User Interface



This screenshot shows the same "Quiz Generator" interface as Fig 5.2, but with data entered into the form fields. The "Select Mode" dropdown remains "MCQ Generation". The "Upload File (PDF or TXT)" section now shows a "Choose File" button and the text "Data.txt". The "Number of MCQs:" field contains the number "5". The "Subject:" field contains "ML". The "Complexity Level:" field contains "Easy". The blue "Submit" button is still present. Below the button, the text "Processing..." is visible.

Select Mode:

Take a Quiz

Upload File (PDF or TXT):

Choose File

Data.txt

Number of MCQs:

5

Subject:

ML

Complexity Level:

Easy

Submit

Processing...

Quiz Generator

Select Mode:

Take a Quiz

Upload File (PDF or TXT):

Choose File

Data.txt

Number of MCQs:

5

Subject:

ML

Complexity Level:

hard

Submit

Submit

Generated MCQs

1. **Machine learning is a part of which field?**
Options:
A) Data Science
B) Artificial Intelligence (AI)
C) Big Data
D) Computer Vision
Correct Answer: Artificial Intelligence (AI)
2. **Which of the following is NOT a recognized type of machine learning?**
Options:
A) Supervised Learning
B) Unsupervised Learning
C) Reinforcement Learning
D) Quantum Learning
Correct Answer: Quantum Learning
3. **Which of the following is a commonly used example of a supervised learning algorithm?**
Options:
A) K-Means Clustering
B) Linear Regression
C) Apriori Algorithm
D) DBSCAN
Correct Answer: Linear Regression

Correct Answer: Linear Regression

4. **What is the main objective of reinforcement learning in machine learning?**
Options:
A) To find patterns in unlabeled data
B) To learn through trial and error to maximize rewards
C) To predict continuous numerical values
D) To classify data into predefined categories
Correct Answer: To learn through trial and error to maximize rewards
5. **Which of the following are common steps in the machine learning workflow?**
Options:
A) Data Cleaning
B) Feature Scaling
C) Data Preprocessing
D) All of the above
Correct Answer: All of the above

Complexity Analysis

The questions range from basic to moderate complexity, covering foundational concepts and specific details.

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1. Which of the following is NOT an essential step in the machine learning pipeline?

Options:

- ☐ A) Data Cleaning
- ☐ B) Feature Scaling
- ☒ C) Model Deployment
- ☐ D) Data Preprocessing

2. Which supervised learning algorithm is best suited for predicting continuous numerical values?

Options:

- ☐ A) Logistic Regression
- ☐ B) Decision Trees
- ☒ C) Linear Regression
- ☐ D) Naive Bayes

3. Which type of machine learning uses a mix of labeled and unlabeled data?

Options:

- ☐ A) Supervised Learning
- ☐ B) Unsupervised Learning
- ☐ C) Reinforcement Learning
- ☒ D) Semi-Supervised Learning

☐ E) Deep Reinforcement Learning

☐ C) Reinforcement Learning

☒ D) Semi-Supervised Learning

4. Which algorithm is particularly effective in handling non-linear datasets?

Options:

- ☐ A) Linear Regression
- ☒ B) Logistic Regression
- ☐ C) Support Vector Machines (SVM)
- ☐ D) k-Nearest Neighbors (k-NN)

5. Which variant of Naive Bayes is based on binary features?

Options:

- ☐ A) Gaussian Naive Bayes
- ☐ B) Multinomial Naive Bayes
- ☒ C) Bernoulli Naive Bayes
- ☐ D) Decision Trees

Submit All Answers

- ☒ B) To find patterns in unlabeled data
- ☐ C) To use a mix of labeled and unlabeled data for training
- ☐ D) To create its own labels from raw data

Submit All Answers

Your Score: 3/5

Feedback

- **Question:** Which of the following algorithms is used for both classification and regression tasks?
Your Answer: Linear Regression
Correct Answer: Support Vector Machines (SVMs)
Explanation: Your answer is incorrect. Please review the topic.
Review Topic: Review the concept of supervised learning and its algorithms.
- **Question:** What is the primary purpose of Semi-Supervised Learning?
Your Answer: To find patterns in unlabeled data
Correct Answer: To use a mix of labeled and unlabeled data for training
Explanation: Your answer is incorrect. Please review the topic.
Review Topic: Review the concept of supervised learning and its algorithms.

4. Which of the following algorithms is used for both classification and regression tasks?

Options:

- ☐ A) Linear Regression
- ☐ B) Logistic Regression
- ☐ C) Decision Trees
- ☒ D) Support Vector Machines (SVMs)

5. What is the primary purpose of Semi-Supervised Learning?

Options:

- ☐ A) To learn from fully labeled data
- ☐ B) To find patterns in unlabeled data
- ☒ C) To use a mix of labeled and unlabeled data for training
- ☐ D) To create its own labels from raw data

Submit All Answers

Your Score: 5/5

Congratulations! All your answers are correct.

CHAPTER 6 PROJECT 1: IMPLEMENTATION

6.1 IMPLEMENTATION PLATFORM

- **Visual Studio**

Visual Studio is a comprehensive integrated development environment (IDE) developed by Microsoft. It offers advanced features for code editing, debugging, project management, version control integration, testing, and performance profiling. Visual Studio is available in different editions, including Community, Professional, and Enterprise, catering to various developer needs. It supports development for multiple platforms and seamlessly integrates with other Microsoft development tools and services, facilitating end-to-end application development and deployment workflows.

The AI-Powered Quiz Generator was developed using a modern and scalable technology stack to ensure efficiency, performance, and ease of use. The following platforms and tools were utilized during the development process:

- **Programming Language:** Python
Python was used as the core programming language due to its powerful libraries and frameworks for AI/ML and backend development.
- **AI Model & Libraries:**
The question generation logic was implemented using transformer-based language models like OpenAI or Hugging Face models. Additional libraries such as transformers, nltk, and spaCy were used for Natural Language Processing tasks.
- **Frontend Technologies:** HTML, CSS, JavaScript
The frontend of the application was developed using standard web technologies—**HTML** for structure, **CSS** for styling, and **JavaScript** for interactivity. This ensures a responsive and visually appealing user interface.
- **Backend & API:** Flask
A lightweight backend using Flask handled quiz generation logic, API calls to the AI model, and interaction with the frontend.
- **Development Environment:**
 - IDE: VS Code
 - OS: Windows
 - Jupyter Notebooks: Used for model testing and experimentation

This combination of platforms and tools ensured smooth development and deployment of the AI-powered quiz generation system, with scalability for future enhancements.

CHAPTER 7 Project 1: TESTING

7.1 STRATEGIES

The test strategy is a formal description of how a software product will be tested. A test strategy is developed for all levels of testing, as required. The test team analyzes the requirements, writes the test strategy and reviews the plan with the project team. The test plan may include test cases, conditions, and the test environment, a list of related tasks, pass/fail criteria and risk assessment.

The purpose of the testing strategy is to define the overall context for the entire testing process. The process is different depending on the specific characteristics of your solution. In many respects, this is the most important part of the testing process, since all future testing decisions will be made within the context of the strategy.

As a programmer, we have to just do a unit testing which is a part of White Box testing. Other type of the testing in each phase of the software is done by testing department. Unit testing begins at the vortex of the spiral and concentrates on each unit (i.e. component) of the software as implemented in source code

Testing strategies for the **AI-Powered Quiz Generator** ensure the accuracy, usability, and performance of the system, thereby enhancing overall user satisfaction and system reliability. Below are the key testing approaches adopted for this project:

Unit Testing:

Individual modules and functions, such as question generation, answer evaluation, are tested in isolation to verify their correctness. This helps in detecting bugs early during development and ensures each component performs its designated task as expected.

Integration Testing:

Integration testing ensures seamless interaction between components like the AI-based question generation module, the frontend UI, the database, and the result evaluation engine. It validates that modules communicate effectively and the data flow remains intact throughout the system.

Functional Testing:

The functional aspects of the quiz generator—such as quiz creation, question customization, student login, quiz attempt, and result display—are tested to confirm they work according to defined requirements. This ensures that all user interactions perform accurately and deliver the intended results.

User Acceptance Testing (UAT):

End-users, including educators and students, are involved in testing the platform in real-time scenarios. Their feedback is used to validate whether the platform is intuitive, responsive, and meets practical expectations for usability, relevance, and content quality.

Performance Testing:

The system's behavior is evaluated under various loads, especially when multiple users attempt quizzes simultaneously. Performance testing ensures that the AI engine generates quizzes quickly, the system doesn't lag during submission, and results are delivered in a timely manner—providing a smooth and efficient experience for all users.

7.2 TESTING METHODS

To ensure the overall quality and performance of the **AI-Powered Quiz Generator**, various testing methods are adopted during the development and deployment phases. These testing methods help in identifying bugs, ensuring functionality, and improving the user experience. Each method is chosen based on the specific needs of the module being tested.

7.2.1 Unit Testing

Unit testing involves testing each individual component or unit of the quiz generator application. It is the first level of functional testing, conducted during the development phase, with the aim of validating that each unit performs as expected in isolation.

In the context of the quiz generator, a unit could be a specific function such as generating a multiple-choice question, evaluating a student's answer, or fetching quiz data from the database. Unit testing ensures that these components work correctly on their own before integrating them into the full system.

This process is typically carried out using the **white-box testing approach** by developers, where the internal logic of the code is tested. Once the development phase is completed and the software is handed over to test engineers, they continue checking each component independently. This systematic examination of modules is known as **Unit Testing** or **Component Testing** and forms the foundation for building a robust and bug-free application.

7.3 TESTING CASE

Test Case ID	Test Case Description	Input	Expected Result	Actual Result	Status
TC_01	Check quiz generation using given topic	Topic: "Machine Learning"	AI should generate 5–10 relevant questions	As expected	Pass
TC_02	Verify quiz submission functionality	User answers all questions & submits quiz	Score should be displayed correctly	As expected	Pass
TC_04	Check result calculation accuracy	5 correct out of 10	Score should be calculated as 50%	As expected	Pass
TC_05	Test question shuffling/randomization	Generate quiz twice with same topic	Questions should appear in different order	As expected	Pass
TC_06	Test quiz generation with unsupported topic	Topic: "xyz123"	Display message: "Topic not found"	As expected	Pass

Table 7.1 Test Cases

CHAPTER 8 PROJECT 2: INTRODUCTION

8.1 PROJECT SUMMARY

Table 8.1: Project Summary

Project Titles	QA Agent	
Aim	The aim of the AI-Powered QA Agent is to deliver instant, accurate answers to user queries by leveraging AI, NLP, and LLMs. It enables natural language interactions with documents, structured data, supporting real-time retrieval and intelligent reasoning. Designed for professionals, and organizations, this tool enhances information access, decision-making, and productivity by providing context-aware, conversational answers with high reliability.	
Developed At	Sarjen Systems Pvt. Ltd.	
Project Category	Web app	
Tools	IDE	Visual Studio
	Languages/Frameworks	Python, LangGraph, Flask, Streamlit
	Database	ChromaDB
Duration	3 Months (January 2025 to April 2025)	

8.2 PURPOSE

The purpose of the AI-Powered QA Agent is to simplify deviation detection and CAPA generation by enabling users to ask questions in natural language and receive accurate, context-aware answers instantly. By integrating advanced NLP and large language models, the system eliminates the need for manual searching through documents or databases. It is designed to assist users in quickly accessing relevant information, improving efficiency, knowledge discovery, and user experience across educational, professional, and enterprise environments.

8.3 OBJECTIVE

- **Enable natural language querying** to allow users to interact with data sources and documents conversationally.
- **Provide accurate and context-relevant answers** by leveraging state-of-the-art NLP and large language models.
- **Support multiple data formats** such as PDFs, text files, websites, and structured databases.
- **Ensure real-time response generation** to enhance user experience and support quick decision-making.
- **Reduce manual search effort** by automating the process of information retrieval and summarization.
- **Enhance accessibility and usability** for educators, students, professionals, and organizational teams.
- **Maintain data privacy and security** while handling sensitive or proprietary information.

8.4 SCOPE

The AI-Powered QA Agent is designed to interpret user queries in natural language and retrieve precise answers from various sources such as documents (PDFs, Word files), websites, and structured databases. The system will utilize advanced NLP techniques and large language models to understand context, extract relevant information, and generate concise, reliable responses.

The scope includes:

- Integration with multiple data sources (PDFs, text files, websites, databases).
- Real-time question processing and answer generation.
- Support for both factual and context-based queries.
- User-friendly chat interface for seamless interaction.
- Admin panel for uploading and managing knowledge base documents.
- Basic analytics to monitor question trends and system performance.

The scope does **not** include:

- Generating answers from external sources not included in the uploaded data (unless explicitly integrated).
- Performing advanced analytics or visualizations beyond textual Q&A.
- Real-time web browsing or search (unless connected to a live search API).

8.5 TECHNOLOGY AND LITERATURE REVIEW

Front End	Streamlit
Back End	LangGraph, RAG
Database	ChromaDB

Table 8.2 Technologies Used to Develop Project

8.6 PLANNING

8.6.1 Project / Internship Development Approach and Justification

The AI-Powered QA Agent will follow an **Agile development approach** to ensure adaptability, continuous feedback, and iterative enhancements throughout the development cycle. The project will be structured using **modular development**, dividing it into distinct components such as data ingestion, embedding generation, retrieval mechanisms, answer generation, frontend UI, and backend API. This modularity ensures streamlined development, simplified debugging, and easier scalability.

A **user-centered design philosophy** will guide the development of the interface, focusing on intuitive interaction, natural language support, and minimal learning curve for users such as students, researchers, and enterprise teams. Key technologies include **Large Language Models (LLMs)** for natural language understanding and answer generation, **Retrieval-Augmented Generation (RAG)** for accurate context-based responses, and **vector databases** like FAISS or Chroma for semantic search efficiency.

Cloud deployment solutions will be leveraged to ensure scalability, high availability, and cost-efficiency. Regular **usability testing** and **feedback collection** will be conducted to iteratively refine the system based on real-world usage and user expectations.

This approach guarantees a flexible, scalable, and high-performance system that aligns with modern AI development practices. The choice of cutting-edge technologies like LLMs, RAG, and cloud deployment is justified by their proven effectiveness in delivering intelligent, responsive, and scalable solutions. The Agile methodology, combined with modular design and user-focused development, ensures that the QA Agent is built to be robust, maintainable, and aligned with user needs—while remaining open to future improvements and innovations.

8.6.2 Roles and Responsibilities

1. Project Manager

- **Responsibilities:**
 - Oversee the entire project lifecycle, ensuring it stays on track, within budget, and meets deadlines.
 - Coordinate between stakeholders, team members, and clients.
 - Manage project risks, resolve issues, and ensure that all tasks are completed on time.
 - Monitor project progress and conduct regular reviews with the team.

2. AI/ML Engineer

- **Responsibilities:**
 - Design and develop AI models, including **LLMs (Large Language Models)**, for question generation.
 - Train, test, and fine-tune AI models using relevant datasets to improve question quality and relevance.
 - Integrate **NLP (Natural Language Processing)** techniques for better understanding of content and context.
 - Work with the development team to integrate AI functionalities into the backend

CHAPTER 9 PROJECT 2: SYSTEM ANALYSIS

9.1 STUDY OF CURRENT SYSTEM

Currently, information retrieval from documents, websites, or databases typically relies on **manual search methods**, including keyword-based searches, control+F navigation, or browsing through structured data using filters or SQL queries. These methods are often **time-consuming, inefficient**, and require users to have prior knowledge of the content layout or data schema.

Some organizations use **basic chatbots or rule-based systems**, which are limited in their ability to understand complex natural language queries or provide contextually accurate answers. These systems often fail to adapt to new information unless manually reprogrammed and do not support real-time integration with dynamic or unstructured data sources like PDFs or web pages.

Additionally, most existing systems lack the ability to:

- Interpret queries in **natural language**.
- Retrieve **context-aware, concise answers** from diverse data sources.
- Integrate **advanced AI techniques** like LLMs and semantic search.
- Provide a **conversational interface** that mimics human-like Q&A interaction.

This gap highlights the need for an intelligent, AI-powered QA solution that can **understand natural language, retrieve relevant answers efficiently**, and **adapt to evolving data**, thereby enhancing user experience and productivity.

9.2 WEAKNESS OF CURRENT SYSTEM

The primary weaknesses of the current QA systems include:

Manual Search Dependency: Users must spend significant time scanning documents or databases to find relevant information, leading to inefficiency and frustration.

Poor Natural Language Understanding: Traditional systems rely on exact keyword matches and cannot interpret the intent or context of user queries effectively.

Lack of Context Awareness: Most systems cannot handle follow-up questions or maintain conversational flow, resulting in fragmented user experiences.

Limited Support for Unstructured Data: Existing tools struggle to extract answers from unstructured formats like PDFs, articles, or web pages.

No Real-Time Intelligence: These systems do not leverage modern AI or retrieval techniques like RAG, limiting their ability to provide dynamic, accurate answers.

Low Engagement: Without interactive, conversational interfaces, user experience is often static and unengaging.

9.3. REQUIREMENT OF NEW SYSTEM

To overcome the limitations of existing systems, the new AI-Powered QA Agent must meet the following requirements:

Natural Language Processing (NLP):

The system should understand and process user queries in natural language to deliver accurate and relevant answers.

Contextual Awareness:

It must maintain conversation history to handle follow-up questions and provide coherent, context-aware responses.

Multi-format Data Support:

The system should be able to extract information from various data sources, including PDFs, text files, websites, and structured databases.

Retrieval-Augmented Generation (RAG):

Implementing RAG architecture will allow the system to fetch relevant data segments before generating answers using LLMs.

Real-time Response Generation:

Users should receive instant answers with minimal latency for a seamless experience.

User-friendly Interface:

A simple and intuitive chat-based interface should be provided for both technical and non-technical users.

Scalability and Cloud Deployment:

The system must support deployment on cloud platforms to ensure scalability, accessibility, and cost efficiency.

Security and Privacy:

It should ensure secure handling of sensitive documents and protect user data.

Feedback and Learning Mechanism:

Incorporate a feedback system to learn from user interactions and continuously improve answer quality.

9.4. FEATURE OF NEW SYSTEM

Key features of the AI-Powered QA Agent:

Natural Language Understanding (NLU):

The system will comprehend user queries in natural language, allowing users to ask questions in conversational language without needing to follow strict syntax.

Context-Aware Responses:

It will maintain context throughout a session, handling follow-up questions and ensuring relevant and coherent answers based on previous interactions.

Multi-Source Data Integration:

The system will support various data formats (PDFs, text files, websites, databases), enabling flexible information retrieval from both structured and unstructured sources.

Real-Time Answer Generation:

Using advanced AI techniques like Retrieval-Augmented Generation (RAG), it will fetch relevant data and provide answers instantly, enhancing user experience and engagement.

Customizable Question Types:

For use cases like education or enterprise knowledge bases, the system will allow customization of question formats (e.g., multiple choice, true/false, open-ended).

Interactive Chat Interface:

A user-friendly interface will allow for easy interaction with the system, enabling

natural conversational exchanges and simplifying user engagement.

Scalability and Cloud Deployment:

The system will be cloud-based, ensuring scalability, high availability, and the ability to handle a growing number of queries without compromising performance.

Continuous Learning and Adaptation:

The system will include a feedback loop, allowing it to learn from user interactions and improve the quality of responses over time.

Security and Data Privacy:

The system will ensure secure handling of sensitive data, with encryption and robust privacy measures to protect user information.

Multi-Language Support (Optional):

The system may support multiple languages, enabling users to interact in their preferred language for greater accessibility.

9.5. SELECTION OF HARDWARE AND SOFTWARE

- Hardware
 - Development Machine Requirement (Average performance)
 - 8 GB RAM
 - 50 GB HDD
 - Client Machine Requirement
 - 4 GB RAM
 - 5 GB of HDD space
 - Host Machine Requirement
 - 32 GB RAM
 - 500GB HDD Space
- Software
 - Development Machine Requirements
 - Visual Studio
 - Client Machine Requirements
 - Browser

CHAPTER 10 PROJECT 2: SYSTEM DESIGN

10.1. FLOWCHART

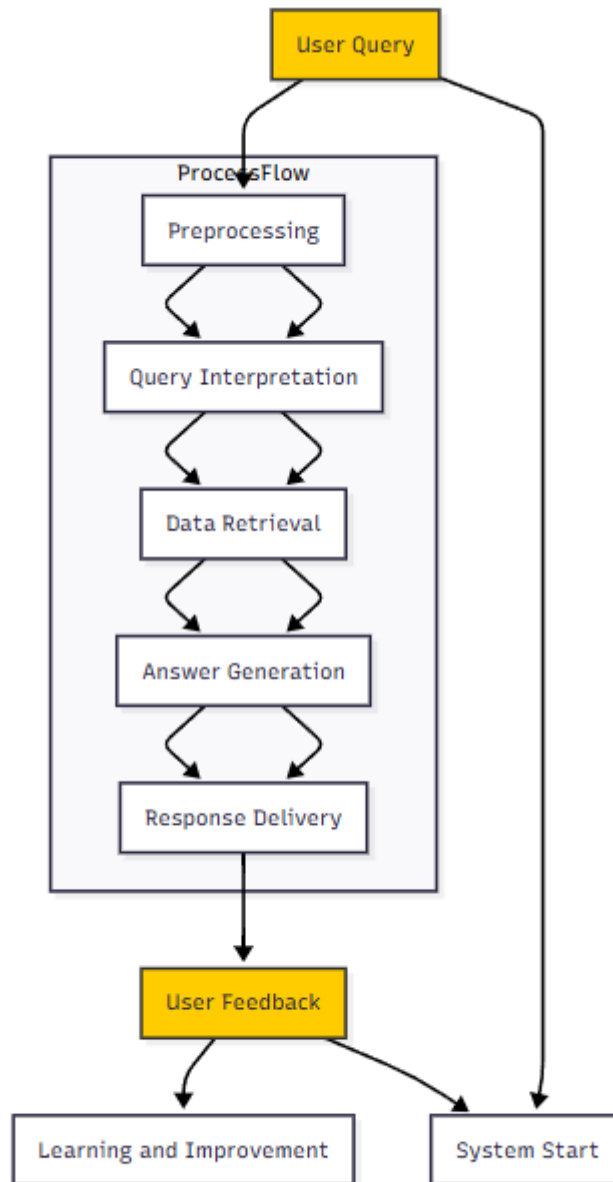
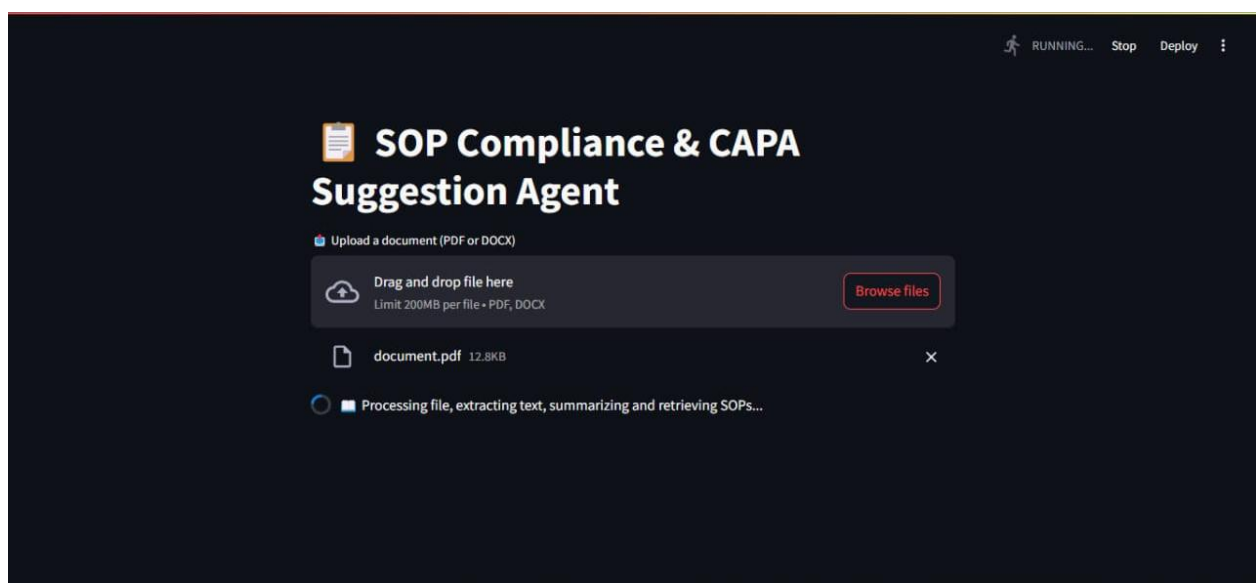


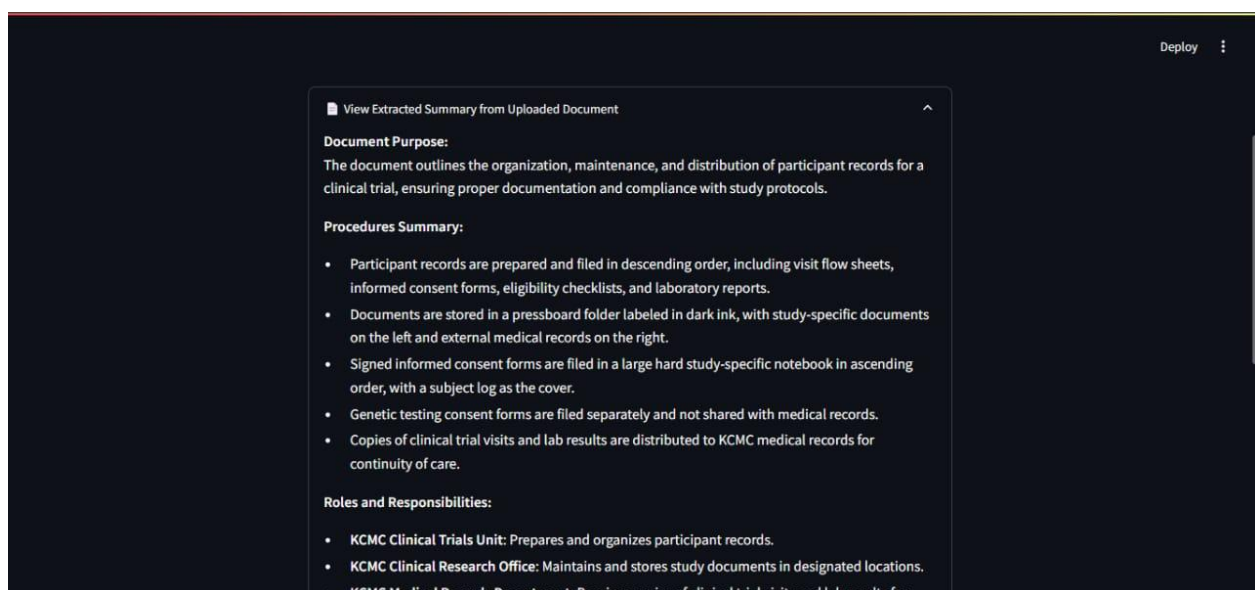
Fig 10.1 Flowchart

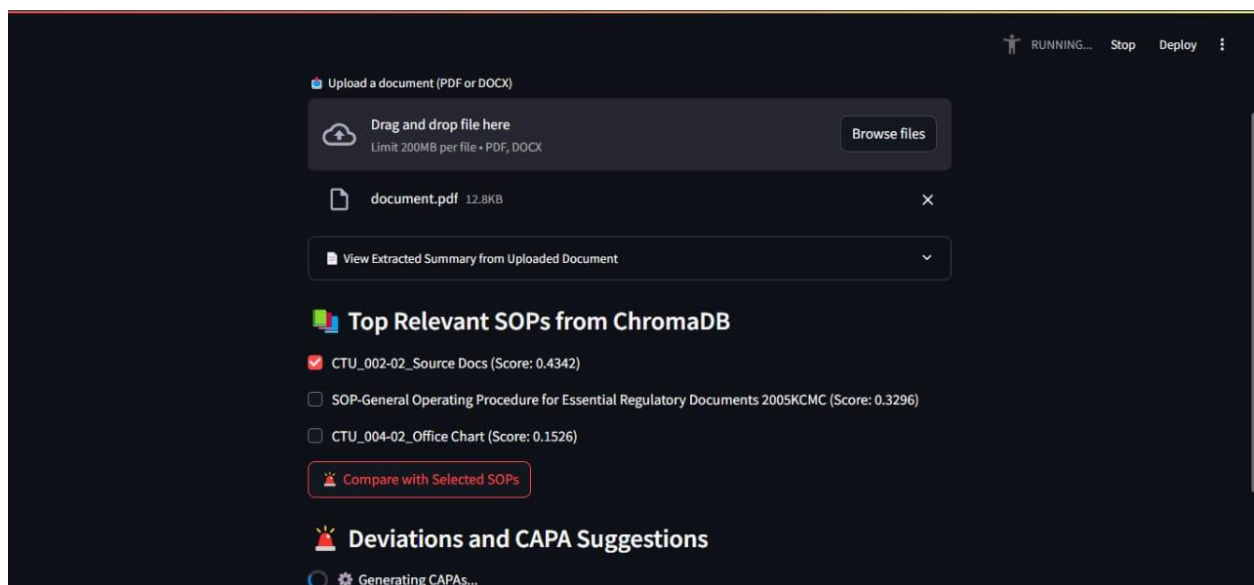
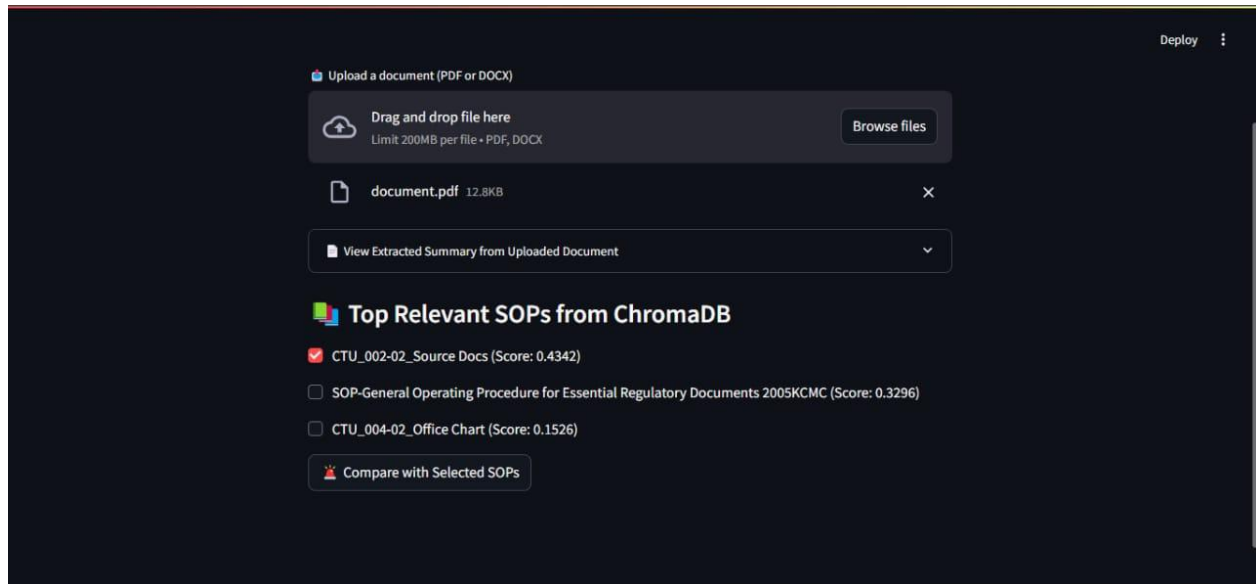
10.2. INTERFACE DESIGN



Fig 10.2 User Interface







Deploy

!

Deviation 1

- SOP Section: Scope
- Content: No mention of compliance with GCP or CFR.
- Impact: Lack of regulatory compliance could lead to non-conformance with standards, affecting audit trails and data integrity.

✂

CAPA Suggestion

▼

{

"root_cause" :

"The SOP was not updated to include references to GCP and CFR, leading to a lack of compliance guidance."

"corrective_action" :

"Update the SOP's Scope section to explicitly mention compliance with GCP and CFR. Conduct staff training on the revised SOP."

"preventive_action" :

"Implement a periodic review process for SOPs to ensure they remain current with regulatory requirements. Provide ongoing training on regulatory standards for all relevant staff."

"risk_level" : "Medium"

}

Deploy

!

Deviation 1

- SOP Section: Scope
- Content: No mention of compliance with GCP or CFR.
- Impact: Lack of regulatory compliance could lead to non-conformance with standards, affecting audit trails and data integrity.

✂

CAPA Suggestion

▶

{ ... }

!

Deviation 2

- SOP Section: Key Procedures - Addenda Process
- Content: No description of correction procedures using addenda.
- Impact: Without proper correction methods, data integrity and traceability may be compromised.

✂

CAPA Suggestion

▼

{

Deploy :

! Deviation 4

- **SOP Section:** Compliance Notes
- **Content:** No mention of adherence to local policies or IRB/IEC.
- **Impact:** Non-compliance with institutional policies could result in regulatory issues.

✂ CAPA Suggestion

```
{
  "root_cause" :
    "Omission of regulatory compliance elements in SOP documentation"
  "corrective_action" :
    "Update SOP to include references to local policies and IRB/IEC requirements"
  "preventive_action" :
    "Implement a review checklist for SOP development to ensure compliance elements
    are included"
  "risk_level" : "Medium"
}
```

Download Deviation + CAPA Report (PDF)

CHAPTER 11 PROJECT 2: IMPLEMENTATION

11.1 IMPLEMENTATION PLATFORM

- **Visual Studio**

Visual Studio is a comprehensive integrated development environment (IDE) developed by Microsoft. It offers advanced features for code editing, debugging, project management, version control integration, testing, and performance profiling. Visual Studio is available in different editions, including Community, Professional, and Enterprise, catering to various developer needs. It supports development for multiple platforms and seamlessly integrates with other Microsoft development tools and services, facilitating end-to-end application development and deployment workflows.

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- **Programming Language:** Python
Python was used as the core programming language due to its powerful libraries and frameworks for AI/ML and backend development.
- **AI Model & Libraries:**
The question generation logic was implemented using transformer-based language models like OpenAI or Hugging Face models. Additional libraries such as `transformers`, `nltk`, and `spaCy` were used for Natural Language Processing tasks.
- **Frontend Technologies:** Streamlit
The frontend of the AI-Powered QA Agent is developed using **Streamlit**, a Python-based framework designed for building interactive web applications. Streamlit allows for quick prototyping and development of intuitive and responsive interfaces, ensuring seamless integration with the backend while providing users with a clean and engaging interface for querying the system. Its simplicity and flexibility make it ideal for building data-driven applications with minimal coding overhead.
- **Development Environment:**
 - IDE: VS Code
 - OS: Windows
 - Jupyter Notebooks: Used for model testing and experimentation

This combination of platforms and tools ensured smooth development and deployment of the AI-powered quiz generation system, with scalability for future enhancements.

CHAPTER 12 PROJECT 2: TESTING

12.1 STRATEGIES

The test strategy is a formal description of how a software product will be tested. A test strategy is developed for all levels of testing, as required. The test team analyzes the requirements, writes the test strategy and reviews the plan with the project team. The test plan may include test cases, conditions, and the test environment, a list of related tasks, pass/fail criteria and risk assessment.

The purpose of the testing strategy is to define the overall context for the entire testing process. The process is different depending on the specific characteristics of your solution. In many respects, this is the most important part of the testing process, since all future testing decisions will be made within the context of the strategy.

As a programmer, we have to just do a unit testing which is a part of White Box testing. Other type of the testing in each phase of the software is done by testing department. Unit testing begins at the vortex of the spiral and concentrates on each unit (i.e. component) of the software as implemented in source code

Testing strategies for the **AI-Powered Quiz Generator** ensure the accuracy, usability, and performance of the system, thereby enhancing overall user satisfaction and system reliability. Below are the key testing approaches adopted for this project:

Unit Testing:

Individual modules and functions, such as question generation, answer evaluation, are tested in isolation to verify their correctness. This helps in detecting bugs early during development and ensures each component performs its designated task as expected.

Integration Testing:

Integration testing ensures seamless interaction between components like the AI-based question generation module, the frontend UI, the database, and the result evaluation engine. It validates that modules communicate effectively and the data flow remains intact throughout the system.

Functional Testing:

The functional aspects of the quiz generator—such as quiz creation, question customization, student login, quiz attempt, and result display—are tested to confirm they work according to defined requirements. This ensures that all user interactions perform accurately and deliver the intended results.

User Acceptance Testing (UAT):

End-users, including educators and students, are involved in testing the platform in real-time scenarios. Their feedback is used to validate whether the platform is intuitive, responsive, and meets practical expectations for usability, relevance, and content quality.

Performance Testing:

The system's behavior is evaluated under various loads, especially when multiple users attempt quizzes simultaneously. Performance testing ensures that the AI engine generates quizzes quickly, the system doesn't lag during submission, and results are delivered in a timely manner—providing a smooth and efficient experience for all users.

12.2 TESTING METHODS

To ensure the overall quality, performance, and reliability of the AI-Powered QA Agent, various testing methods will be employed during the development and deployment phases. These methods will help identify bugs, ensure functionality, and enhance the user experience. Each testing method will be chosen based on the specific requirements of the module being tested.

12.2.1 Unit Testing

Unit testing involves testing each individual component or unit of the QA Agent application. This is the first level of functional testing, carried out during the development phase to ensure that each unit functions as expected in isolation.

In the context of the QA Agent, a unit could be a specific function, such as:

- Interpreting a user query,
- Fetching relevant data from a database or external source,
- Generating an accurate answer based on retrieved information.

Unit testing ensures that these individual components perform their tasks correctly before integrating them into the entire system.

This testing is typically done using a white-box testing approach by developers, who test the internal logic of the code. After the development phase is complete, test engineers continue verifying each component independently. This structured approach helps in identifying and fixing issues at the module level, ensuring that the QA Agent is free from bugs and performs reliably when all components are integrated.

12.3 TESTING CASE

Test Case ID	Test Case Description	Input	Expected Result	Actual Result	Status
TC_01	Check deviation detection accuracy	Input deviation scenario	System should flag deviation	As expected	Pass
TC_02	Verify CAPA generation for detected deviation	Deviation detected, generate CAPA	Valid CAPA including corrective actions	As expected	Pass
TC_03	Validate CAPA effectiveness	Implement CAPA, monitor results	No recurrence of deviation	As expected	Pass
TC_04	Check handling of false positives/negatives	Input normal and deviation data	Only true deviations should be flagged	As expected	Pass
TC_05	Test system response time for deviation detection	Submit deviation, measure response time	Detection and CAPA generation within 30-40 seconds	As expected	Pass

Table 12.1 Test Cases

CHAPTER 13: CONCLUSION AND DISCUSSION

13.1 OVERALL ANALYSIS OF INTERNSHIP

The internship experience provided invaluable insights into real-world software development, project life cycles, and the application of emerging technologies. During this period, I worked on two key projects: the **AI-Powered Quiz Generator** and the **AI-Powered QA Agent**, both of which allowed me to deepen my technical skills and enhance my problem-solving abilities.

Working on the **AI-Powered Quiz Generator** helped me apply my academic knowledge to practical challenges, particularly in the areas of Python development, AI-driven question generation, and user interface design. I gained hands-on experience with modern tools for version control, project collaboration, and system testing. This project enhanced my skills in natural language processing (NLP) and machine learning, particularly in how AI can be used for educational purposes. I also improved my ability to work in teams, communicate effectively, and integrate feedback during iterative development.

In parallel, the **AI-Powered QA Agent** project gave me a deeper understanding of deviation detection and CAPA (Corrective and Preventive Actions) generation. I gained valuable experience in integrating AI with real-world processes for error detection, risk management, and process optimization. This project also provided insight into how AI can be applied in quality assurance, and I learned how to design and test complex systems that automate decision-making processes.

Both projects allowed me to further develop my skills in Python, AI, machine learning, system integration, and testing strategies. They also exposed me to industry-standard practices like Agile development and iterative testing. Overall, this internship not only strengthened my technical expertise but also sharpened my soft skills, preparing me for future professional roles in the software development and AI fields.

13.2 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

Throughout my internship, I encountered several challenges while working on the **AI-Powered Quiz Generator** and **AI-Powered QA Agent** projects. These challenges provided opportunities for learning and growth, and I was able to devise solutions to overcome them.

AI-Powered Quiz Generator

1. Problem: Inconsistent Question Generation Quality

- While implementing the AI-driven question generation, the output was inconsistent, with some questions lacking relevance or clarity.
- **Solution:** I fine-tuned the AI model by adding domain-specific training data to improve the quality and relevance of generated questions. Additionally, I incorporated multiple layers of validation, such as verifying the question's structure and its alignment with the input topic.

2. **Problem: Handling Unsupported Topics**

- The system faced issues when generating quizzes for unsupported or vague topics, resulting in errors or empty outputs.
- **Solution:** I implemented a fallback mechanism where the system checks the topic's validity before attempting to generate questions. If a topic is unsupported, a user-friendly message is displayed, preventing system crashes and improving user experience.

3. **Problem: User Interface Responsiveness**

- The front-end design faced occasional responsiveness issues on mobile devices.
- **Solution:** I worked on refining the CSS layout and incorporated responsive design principles, ensuring a seamless user experience across devices.

AI-Powered QA Agent

1. **Problem: Deviation Detection Accuracy**

- During the initial phase, the system sometimes flagged normal data as deviations, leading to false positives.
- **Solution:** I enhanced the data preprocessing phase and adjusted the detection algorithm to focus more on contextual analysis. This significantly reduced false positives by incorporating a more sophisticated evaluation of the deviation patterns.

2. **Problem: CAPA Generation Reliability**

- The initial CAPA generation often failed to propose relevant corrective actions, especially for complex deviations.
- **Solution:** I introduced a feedback loop to learn from the generated CAPAs and incorporated external data sources to enrich the system's decision-making process. This helped the system generate more accurate and actionable CAPA suggestions.

3. **Problem: System Integration and Scalability**

- Integrating the deviation detection module with the CAPA generation system posed challenges in terms of scalability, especially when processing large datasets.
- **Solution:** I optimized the integration flow by using more efficient data structures and cloud-based storage solutions, ensuring scalability and smooth performance even with large datasets.

13.3 DATE OF CONTINUES EVALUATION

- Feb 17, 2025
- March 16, 2025
- April 20, 2025

13.4 LIMITATIONS

During the development of both the **AI-Powered Quiz Generator** and **AI-Powered QA Agent**, several limitations were encountered. While the systems were designed to address

specific challenges effectively, there are still areas where improvements or future enhancements could be made.

AI-Powered Quiz Generator

1. Limited Topic Scope

- The system relies on predefined datasets and topic-specific training data. While it handles common topics well, it may struggle with generating high-quality questions for niche or less common subjects.
- **Limitation:** The system's effectiveness is limited to topics with sufficient data and predefined models.

2. Quality of Generated Questions

- Although AI can generate a wide range of questions, the quality and relevance of some questions may vary. Complex or ambiguous topics sometimes result in poorly formulated or irrelevant questions.
- **Limitation:** Continuous fine-tuning of the AI model and inclusion of more diverse datasets would be required to improve question quality further.

3. User Customization

- While the system offers a basic set of customization features (e.g., question difficulty), the user experience could be enhanced by offering more advanced features like personalized quiz creation based on student progress.
- **Limitation:** Limited customization options for educators or trainers, especially in terms of dynamically adjusting quizzes to individual learning needs.

AI-Powered QA Agent

1. Deviation Detection Accuracy

- The system, although optimized, may still produce occasional false positives or negatives, particularly in ambiguous or complex scenarios where patterns are hard to define.
- **Limitation:** The detection algorithm needs further refinement to handle edge cases with greater accuracy.

2. Complexity of CAPA Generation

- The CAPA generation process works well for simple deviations but can become less effective when dealing with highly complex, multi-faceted issues. The system might not always generate optimal corrective and preventive actions.
- **Limitation:** The complexity of real-world issues may limit the system's ability to generate highly specific CAPA solutions in all cases.

3. Integration with External Systems

- While the QA agent works effectively in standalone mode, integrating with larger enterprise systems or multiple data sources poses challenges in terms of synchronization and real-time updates.
- **Limitation:** Integration with legacy systems or multiple external data sources might require additional customization and result in higher maintenance overhead

13.5 FUTURE ENHANCEMENT

While the **AI-Powered Quiz Generator** and **AI-Powered QA Agent** projects have successfully addressed key challenges and provided value, there are several areas where enhancements could be implemented to further improve their functionality, scalability, and user experience.

AI-Powered Quiz Generator

1. Increased Topic Coverage

- The quiz generator currently supports a limited range of topics. Future enhancements could include expanding the dataset to cover more niche or specialized subjects, making it more adaptable to diverse user needs.
- **Enhancement:** Integrating APIs or using web scraping techniques to automatically pull in content from various online sources to diversify the available topics.

2. Personalized Quiz Creation

- Future versions could include more advanced algorithms to analyze a learner's performance and generate quizzes tailored to their learning needs and progress.
- **Enhancement:** Implementing machine learning models that analyze individual learning patterns and suggest quiz topics or question types based on past performance.

3. Advanced Customization Options

- The current customization options are basic and could be expanded to allow more control over the quiz format, including options for time limits, multimedia questions (e.g., images, audio), and real-time feedback during quizzes.
- **Enhancement:** Adding interactive elements like drag-and-drop questions, timed quizzes, and question types beyond just multiple-choice.

4. Multilingual Support

- The quiz generator can be expanded to support multiple languages, catering to a broader, global audience.
- **Enhancement:** Implementing NLP models for multilingual question generation, translation of questions, and evaluation of answers in various languages.

AI-Powered QA Agent

1. Improved Deviation Detection Algorithms

- Although the deviation detection works well, it can be improved by using advanced techniques such as deep learning for more accurate anomaly detection, especially in complex data sets.
- **Enhancement:** Integrating advanced deep learning models like autoencoders or LSTM networks to enhance the system's ability to detect subtle deviations.

2. Context-Aware CAPA Generation

- Future enhancements could focus on making the CAPA generation process more context-aware, allowing it to understand the broader implications of deviations and generate more nuanced corrective actions.
- **Enhancement:** Incorporating contextual reasoning and knowledge graphs to provide more effective and customized CAPA solutions based on historical data and industry best practices.

3. **Integration with Other Enterprise Systems**

- Enhancing the integration capabilities of the QA agent with other enterprise tools like CRM systems, databases, and business intelligence platforms would improve its utility in larger organizational environments.
- **Enhancement:** Developing APIs or connectors to integrate seamlessly with existing enterprise systems, ensuring a more holistic approach to quality assurance and process optimization.

4. **Real-Time Monitoring and Alerts**

- Future versions of the QA agent could include real-time monitoring of systems and the ability to provide instant alerts for deviations, enhancing decision-making speed.
- **Enhancement:** Adding real-time analytics and notification systems, where deviations trigger immediate alerts to the relevant stakeholders for faster response.

5. **Predictive Analytics for CAPA Effectiveness**

- Predictive analytics could be implemented to analyze the success rates of previously generated CAPAs, providing insights into what corrective actions are most effective in different contexts.
- **Enhancement:** Implementing machine learning models to predict the likelihood of CAPA success based on historical data and offer recommendations for improvements.

Topics	References
LangChain	https://python.langchain.com/docs/introduction/
Groq	https://console.groq.com/docs/overview
Streamlit	https://docs.streamlit.io/
LangGraph	https://langchain-ai.github.io/langgraph/
Agentic Framework LangGraph	https://medium.com/@lorevanoudenhove/how-to-build-ai-agents-with-langgraph-a-step-by-step-guide-5d84d9c7e832
LangSmith	https://docs.smith.langchain.com/
ChromaDB	https://docs.trychroma.com/docs/overview/introduction
RAG Architecture	https://medium.com/@tejpal.abhyuday/retrieval-augmented-generation-rag-from-basics-to-advanced-a2b068fd576c
Flask	https://flask.palletsprojects.com/en/stable/

Table 13.1 References